PADHAKU
Contemporary India · Class 10 Social Science



CHAPTER 1

Resources and Development

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🧪 Activity · Resources

? Can you identify and name the various items used in making life comfortable in our villages and towns? List the items and name the material used in their making.

💡 SAMPLE ANSWER

Item	Material used
Chair, table, door	Wood (from forests)
Utensils	Steel, aluminium, copper (minerals)
House	Bricks (clay), cement (limestone), iron rods, sand
Clothes	Cotton (plants), wool (sheep), polyester (petroleum)
Bucket, pipes	Plastic (petroleum)
Electric wires, fan	Copper, aluminium, iron
Window glass	Sand (silica)

All these materials come from our environment and become **resources** when we use them with technology.

? Do you think that resources are free gifts of nature as is assumed by many?

✔ ANSWER

No. Resources are not free gifts of nature.

- Resources are a **function of human activities**.
- Human beings use technology to change things in nature into useful resources (e.g. a river becomes a resource when a dam produces electricity).
- Human beings themselves are essential parts of resources.
- A thing becomes a resource only when it is technologically accessible, economically feasible and culturally acceptable.

🧪 Activity · Development of Resources

1 **Imagine, if the oil supply gets exhausted one day, how would this affect our life style?**

💡 SAMPLE ANSWER

- **Transport** would stop: cars, buses, trucks, tractors and aeroplanes mostly run on petrol and diesel.
- **Farming** would suffer: tractors, pumps and harvesters need diesel, so food production would fall.
- Prices of food and goods would rise because goods could not be moved easily.
- Many **industries** (plastics, paints, fertilisers, medicines) use petroleum and would close; many people would lose jobs.
- Cooking gas (LPG) and kerosene would not be available.

- We would have to depend on bicycles, walking, electric vehicles and renewable energy such as solar and wind power.

So we must use oil carefully and shift to renewable sources now.

2 **Plan a survey in your colony/village to investigate people's attitude towards recycling of the domestic/agricultural wastes. Ask questions about:**
(a) What do they think about resources they use?
(b) What is their opinion about the wastes, and its utilisation?
(c) Collage your results.

💡 SAMPLE ANSWER

Model survey plan

- **Sample:** 20 households (10 in the colony/village, 10 farmers).
- **Questions for (a):** Which resources do you use daily (water, LPG, electricity, wood)? Do you think they will last for ever? Do you try to save them?
- **Questions for (b):** Do you separate wet and dry waste? What do you do with plastic, paper and crop stubble? Do you make compost or give waste to kabadiwala? Would you join a recycling programme?
- **Record** answers in a table: Yes / No / Sometimes.

- **(c) Collage:** paste pictures of compost pits, waste bins, recycled items and draw simple bar graphs of the answers on a chart paper.
- **Conclusion:** e.g. most people think resources are unlimited; few separate waste; awareness is needed.

🧪 Activity · Resource Planning

? What resources are being developed in your surroundings by the community/village panchayats/ward level communities with the help of community participation?

💡 SAMPLE ANSWER

Examples (write those seen in your area):

- Village ponds and check dams for **rainwater harvesting**.
- **Tree planting** on roadsides and common land.
- Community **biogas plants** and compost pits.
- Village roads, wells, hand pumps and tanks.
- Common grazing land and community forests protected by the people.
- Solar street lights in villages and parks in towns.

? Can you name some resource rich but economically backward regions and some

resource poor but economically developed regions? Give reasons for such a situation.

💡 SAMPLE ANSWER

Resource rich but economically backward: Jharkhand, Chhattisgarh, Odisha (rich in minerals and coal), Arunachal Pradesh (rich in water resources).

Resource poor but economically developed: e.g. Punjab and Delhi in India; Japan in the world.

Reasons:

- Mere availability of resources does not bring development.
- Development needs proper **technology**, **skilled human resources** and good **institutions**.
- Backward regions lack infrastructure (roads, power, industries), capital and skilled people.
- Developed regions use advanced technology, trade and skilled people to bring resources from elsewhere.
- Historical experience (such as colonisation) also matters.

? **Prepare a list of resources found in your state and also identify the resources that are important but deficit in your state.**

💡 SAMPLE ANSWER

Model answer (for **Rajasthan** – make your own for your state):

- **Resources found:** solar energy, wind energy, minerals (zinc, copper, marble, gypsum, limestone), livestock, tourism (forts, deserts).
- **Important but deficit:** water, forests, fertile soil.

Another example, **Delhi:** found – skilled human resources, industry, trade; deficit – water, land, minerals, forests.

🧪 Activity · Land Use Pattern in India

? Try to do a comparison between the two pie charts (Fig. 1.4) given for land use and find out why the net sown area and the land under forests have changed from 1960-61 to 2019-20 very marginally.

✔ ANSWER

Category	1960–61	2019–20
Forest	18.11%	23.41%
Barren and unculturable waste land	12.01%	5.40%
Area under non-agricultural uses	4.95%	9.06%
Permanent pasture and grazing land	4.71%	3.42%
Area under misc. tree crops and groves	1.50%	1.02%

Culturable waste land	6.23%	3.90%
Fallow other than current fallow	3.50%	3.67%
Current fallow	3.73%	4.49%
Net sown area	45.26%	45.64%

Why net sown area changed very little:

- Almost all good land was already under cultivation in 1960–61.
- New land brought under farming was balanced by farm land lost to houses, roads, railways and industries (non-agricultural uses nearly doubled).

Why forest area changed only a little:

- Growing population needs land for farming, housing, mining, dams and industries, so forests are cut.
- Afforestation has added some area, but it is still far below the desired 33% of the National Forest Policy (1952).

? **The land under permanent pasture has also decreased. How are we able to feed our huge cattle population on this pasture land and what are the consequences of it?**

💡 SAMPLE ANSWER

How cattle are fed: on crop residues (straw, stalks), fodder crops grown on farm land, cattle feed bought from markets, grazing in forests, fallow lands and roadsides.

Consequences:

- **Overgrazing** of the little pasture left, which causes land degradation (e.g. in Gujarat, Rajasthan, Madhya Pradesh, Maharashtra).
- Loss of grass cover leads to soil erosion.
- Cattle are underfed, so milk yield is low.
- Farm land is used for fodder instead of food crops; cost of rearing animals rises.

? **Find out reasons for the low proportion of net sown area in these states (Arunachal Pradesh, Mizoram, Manipur and Andaman Nicobar Islands).**

✔ ANSWER

- **Hilly and mountainous relief** with steep slopes; little flat land for farming.
- Large area under **dense forests**.
- Heavy rainfall causes soil erosion; soils are often thin or acidic.
- **Shifting (jhum) cultivation** is practised, so much land stays fallow.
- Poor transport and irrigation facilities.
- Andaman and Nicobar Islands are small islands, mostly forest-covered, and much area is protected.

📝 Exercises

1 (i) **Which one of the following is the main cause of land degradation in Punjab?**

(a) Intensive cultivation (b) Deforestation (c) Over irrigation (d) Overgrazing

✔ ANSWER

(c) Over irrigation

Over irrigation causes waterlogging, which increases salinity and alkalinity in the soil.

1 (ii) **In which one of the following states is terrace cultivation practised?**

(a) Punjab (b) Plains of Uttar Pradesh (c) Haryana (d) Uttarakhand

✔ ANSWER

(d) Uttarakhand

It is a hilly state (western/central Himalayas), where steps are cut on slopes for farming.

1 (iii) **In which of the following states black soil is predominantly found?**

(a) Uttar Pradesh (b) Maharashtra (c) Rajasthan (d) Jharkhand

✔ ANSWER

(b) Maharashtra

Black soil is typical of the Deccan trap (basalt) region, which covers the plateau of Maharashtra.

2 (i) **Name three states having black soil and the crop which is mainly grown in it.**

✔ ANSWER

- **States:** Maharashtra, Madhya Pradesh and Chhattisgarh (also Gujarat – Saurashtra).
- **Crop: Cotton**. Black soil is ideal for cotton, so it is also called **black cotton soil** (regur soil).

2 (ii) **What type of soil is found in the river deltas of the eastern coast? Give three main features of this type of soil.**

✔ ANSWER

Alluvial soil is found in the deltas of the Mahanadi, Godavari, Krishna and Kaveri.

- It is made of various proportions of **sand, silt and clay**, deposited by rivers.
- It is **very fertile** and contains adequate potash, phosphoric acid and lime.
- It is ideal for **sugarcane, paddy, wheat**, cereals and pulses, so these areas are intensively cultivated and densely populated.

2 (iii) **What steps can be taken to control soil erosion in the hilly areas?**

✔ ANSWER

- **Contour ploughing:** ploughing along contour lines slows the flow of water down the slopes.
- **Terrace farming:** cutting steps on slopes restricts erosion (as in the western and central Himalayas).
- **Strip cropping:** strips of grass grown between crops break the force of wind and water.
- **Afforestation** and control of overgrazing; planting shelter belts of trees.

3 (i) **Explain land use pattern in India and why has the land under forest not increased much since 1960-61?**

✔ ANSWER

The use of land is decided by **physical factors** (topography, climate, soil) and **human factors** (population density, technology, culture and traditions). India's total geographical area is 3.28 million sq km, but land use data is available for only about 93% of it.

Land use pattern (2019–20):

- **Net sown area:** about 45.6%; over 80% in Punjab and Haryana, less than 10% in Arunachal Pradesh, Mizoram, Manipur and Andaman and Nicobar Islands.
- **Forests:** about 23.4%, far below the desired 33% (National Forest Policy, 1952).

- **Barren and waste land:** about 5.4%; **non-agricultural uses** (settlements, roads, railways, industry): about 9.1%.
- **Permanent pastures:** about 3.4% and decreasing.
- **Fallow lands:** about 8% (current fallow 4.49% + other fallow 3.67%); culturable waste land about 3.9%.

Why forest land has not increased much:

- Growing population needs more land for agriculture, housing and roads.
- Forests are cleared for **mining, industries, dams** and other projects (e.g. in Jharkhand, Chhattisgarh, Odisha).
- Overgrazing and shifting cultivation damage forests.
- New plantations only partly make up for the loss.

So forest cover rose only from about 18% (1960–61) to about 23% (2019–20).

3 (ii) **How have technical and economic development led to more consumption of resources?**

✔ ANSWER

Human beings change materials in nature into resources with the help of **technology** and **institutions**. As technology and the economy grew, the use of resources grew very fast.

- **New technology** made it possible to dig deep mines, drill oil wells and cut forests quickly, so more resources could be taken out.
- **Industrialisation and mass production** need huge amounts of raw materials, minerals, water and energy.
- **Better transport** (railways, ships, roads) allowed resources to be moved and traded across the world.
- **Economic growth** raised people's incomes, so they buy more goods (cars, electronics, packaged goods) – more demand means more consumption.
- **Urbanisation** needs land, water, cement and energy for cities.
- Countries with higher technology, as in colonisation, exploited the resources of other regions.

This has led to depletion of resources, their accumulation in few hands and problems like global warming and pollution. Hence **sustainable development** and resource conservation are needed.

🧪 Project/Activity

1 **Make a project showing consumption and conservation of resources in your locality.**

💡 SAMPLE ANSWER

Model project plan

- **Title:** Resources in my locality – use and care.
- **Part 1 – Consumption:** list resources used (water, electricity, LPG, fuel, paper, land); note daily use in 5 homes in a table; add photos.
- **Part 2 – Wastage seen:** leaking taps, lights left on, burning of waste, plastic dumping.
- **Part 3 – Conservation:** rainwater harvesting, solar panels, LED bulbs, composting, recycling, tree planting, using public transport.
- **Conclusion:** suggestions for families and the RWA/panchayat; a chart with the slogan "Reduce, Reuse, Recycle".

2 **Have a discussion in the class – how to conserve various resources used in your school.**

💡 **SAMPLE ANSWER**

Points for the discussion:

- **Electricity:** switch off fans and lights when leaving the class; use LED bulbs; solar panels on the roof.
- **Water:** close taps, fix leaks, rainwater harvesting, use waste water for the garden.
- **Paper:** write on both sides, reuse old notebooks, donate old books, recycle waste paper.

- **Food:** do not waste mid-day meal; make compost from food waste.
- **Furniture and land:** take care of desks; plant trees in the school compound.
- Form an **eco-club** to check and remind everyone.

3 **Imagine if oil supplies get exhausted, how will this affect our life style?**

💡 **SAMPLE ANSWER**

- Vehicles, trains (diesel), ships and aeroplanes would stop; travel would become slow and costly.
- Tractors and pump sets would stop, so food production would fall and prices would rise.
- Industries making plastics, fertilisers, paints and medicines would be badly hit; unemployment would rise.
- No LPG or kerosene for cooking and lighting.
- We would have to shift to solar, wind, hydro power and electric vehicles, and walk or cycle more.

So we must save oil now and develop renewable energy.

4 **Solve the puzzle by following your search horizontally and vertically to find the hidden answers.**

(i) Natural endowments in the form of land, water, vegetation and minerals.

(ii) A type of non-renewable resource.

(iii) Soil with high water retaining capacity.

(iv) Intensively leached soils of the monsoon climate.

(v) Plantation of trees on a large scale to check soil erosion.

(vi) The Great Plains of India are made up of these soils.

✔ ANSWER

Clue	Answer	Where in the grid
(i)	RESOURCE(S)	Vertical, 8th column
(ii)	MINERALS	Horizontal, 6th row
(iii)	BLACK (soil)	Vertical, 1st column
(iv)	LATERITE	Horizontal, 8th row
(v)	AFFORESTATION	Horizontal, 2nd row
(vi)	ALLUVIAL	Vertical, 10th column

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**